# The order of saddle points in tree tensor network training

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**Abstract**

We study gradient flow on tree tensor networks (TTNs) fitted to a target that is a sum of rank-one tensors. From small initialization the flow learns the rank-one terms one at a time and stays near a saddle point between consecutive terms. How long it stays is controlled by the order of the saddle, the exponent of the fastest local decrease of the loss. We compute this order exactly at the critical point that represents one learned term with one channel per edge. Typically it equals the number of cores in the smallest subtree that connects the cores on which some unlearned term differs from the learned one. This number depends on the shape of the tree, so the same target produces plateaus of different lengths on different trees. The order is lower only when the unlearned terms are, in a precise spectral sense, stronger than the learned one, and we give its value in that case too. At leading order, a saddle of order $p$ is left after a time $\propto \ln(1/\sigma)$ if $p = 2$ and $\propto \sigma^{-(p-2)}$ if $p \geq 3$, where $\sigma$ is the initialization scale; when two descent mechanisms of different order coexist, the lower order governs the escape only below a crossover scale. Simulations on nine trees agree with these laws.

## 1 Introduction

In a deep linear network (DLN) trained from small initialization, gradient flow learns the singular modes of the target one at a time, with a plateau before each (Saxe et al., 2014; Jacot et al., 2022). Every plateau lasts $\propto \sigma^{-(K-2)}$, where $K$ is the depth and $\sigma$ the initialization scale. The reason is that the output is a product of $K$ layers that all start at size $\sigma$: their common size $a$ obeys $\dot{a} \approx c\, a^{K-1}$, whose solution from $a(0) = \sigma$ stays small for a time $\propto \sigma^{-(K-2)}$ and then grows quickly. In a DLN every mode needs every layer, so every plateau has the same exponent.

A TTN is also a product of $K$ factors, its cores, but each core is attached to only some of the inputs. After the first term is learned, most cores are large. To add the next term, only some cores need to change from small values, and the length of the next plateau depends on how many. In terms of the loss landscape, it depends on the order of the saddle at which the flow rests.

This note determines that order (Theorem 1) and derives the resulting plateau lengths at leading order (Section 5). Section 6 compares the result with prior work.

## 2 Setting and definitions

**Tree tensor networks.** Let $T$ be a tree with $K$ nodes. Each node $v$ holds a tensor $G_v$, called a *core*. A core has one leg for each tree edge at $v$ and, possibly, one *physical* leg of dimension $D_v$. The legs on a tree edge $e$ have dimension $\chi_e$, the *bond dimension*. Contracting all tree-edge legs gives a tensor $W(\theta) \in \bigotimes_v \mathbb{R}^{D_v}$, where $\theta$ denotes all core entries. The network has $N$ inputs $x_0, \ldots, x_{N-1} \in \mathbb{R}^d$, called *sites*; each site belongs to the physical leg of one core, so $D_v = d^{n_v}$ if core $v$ carries $n_v$ sites ($D_v = 1$ if it carries none). The network computes $f(x) = \langle W, x_0 \otimes \cdots \otimes x_{N-1} \rangle$.

Three trees serve as examples (Figure 1): the matrix product state (MPS), a path with one site per core; the binary tree, with two sites on each bottom core and none on the others; and the hierarchical Tucker (HT) tree, with one site on each leaf and none on the internal cores.

**Target and loss.** Labels are generated by a target tensor $W^*$ as $y = \langle W^*, x_0 \otimes \cdots \otimes x_{N-1} \rangle$. For whitened Gaussian inputs the population squared loss is, up to a constant,

$$L(\theta) = \tfrac{1}{2}\|W(\theta) - W^*\|^2. \tag{1}$$

We take $W^*$ to be a sum of distinct products of standard basis vectors,

$$W^* = \sum_r s_r\, U_r, \qquad U_r = e_{\pi_r(0)} \otimes e_{\pi_r(1)} \otimes \cdots \otimes e_{\pi_r(N-1)}, \qquad s_r > 0, \tag{2}$$

where $\pi_r$ is a string of basis indices, one per site (for example $\pi_r = 110010$). We call $U_r$ the *components* of the target. Since the strings are distinct, the tensors $U_r$ are orthonormal.

**The saddle after one component is learned.** Fix a component $j$ and let $q_v \in \mathbb{R}^{D_v}$ be the basis vector that $U_j$ has on the sites of core $v$, so $U_j = \bigotimes_v q_v$. Let $\theta_S$ have cores

$$G_v = a_v\, q_v \otimes e_0 \otimes \cdots \otimes e_0, \qquad a_v > 0, \qquad \textstyle\prod_v a_v = s_j, \tag{3}$$

where $e_0$ is the first basis vector of each tree-edge leg. Then $W(\theta_S) = s_j U_j$: the network represents component $j$ exactly, using one channel (index 0) on every edge. The other $\chi_e - 1$ indices of each edge are *unused channels*. The loss at $\theta_S$ is $L_S = \frac{1}{2}\|\mathcal{R}\|^2$, with residual $\mathcal{R} = W^* - s_j U_j = \sum_{r \neq j} s_r U_r$; we assume $\mathcal{R} \neq 0$. When some $\chi_e \geq 2$ there are other critical points with $W = s_j U_j$, for example with nonzero entries on an unused channel that meet only zeros in the neighbouring core; they are not of the form (3), and the results below concern the points (3).

Two points of the form (3) differ only in the scalars $a_v$, and they are related by a diagonal change of basis on the edges: scaling index 0 of edge $(u, v)$ by $\lambda$ in $G_u$ and by $1/\lambda$ in $G_v$ leaves $W$ unchanged for every $\theta$ and trades $a_u$ against $a_v$. This map is linear, invertible and preserves $L$, so it preserves the order defined below. We therefore compute at the balanced point $a_v = a := s_j^{1/K}$ where convenient.

**Order of a saddle.** For a critical point $\theta_S$ let $\psi(\tau) = \min_{\|\theta - \theta_S\| \leq \tau} \big(L(\theta) - L_S\big)$. Since $L$ is a polynomial, $\psi$ is semialgebraic, so if $\psi \not\equiv 0$ near 0 then $\psi(\tau) = -c\,\tau^{\alpha}(1 + o(1))$ for some $c > 0$ and rational $\alpha$. The *order* of $\theta_S$ is $\operatorname{ord}(\theta_S) = \alpha$: the exponent of the fastest local decrease of the loss. Equivalently, it is the largest $p$ with $L(\theta) - L_S \geq -C\|\theta - \theta_S\|^p$ near $\theta_S$. (This differs from the derivative-based notion of "higher-order saddle" used, for example, by Anandkumar and Ge, 2016.) At a critical point, order 2 is equivalent to a negative Hessian eigenvalue: if the Hessian is positive semidefinite, Taylor's theorem gives $L - L_S \geq -C\|\theta - \theta_S\|^3$. A saddle of order 2 is called *strict*.

**Steiner subtrees.** For a set $B$ of cores, the *Steiner subtree* $\mathrm{St}(B)$ is the smallest subtree of $T$ containing $B$. We use the symbol both for its set of cores and for the subtree itself: $|\mathrm{St}(B)|$ is its number of cores, and the edges of $\mathrm{St}(B)$ are the tree edges between its cores. For each component $r \neq j$ let

$$V_r = \{\text{cores carrying a site where } \pi_r \text{ and } \pi_j \text{ differ}\}, \qquad k_r = |\mathrm{St}(V_r)|, \qquad k^* = \min_{r \neq j} k_r. \tag{4}$$

Figure 1 gives an example. The point $\theta_S$ is critical iff $k^* \geq 2$ (Lemma 1).

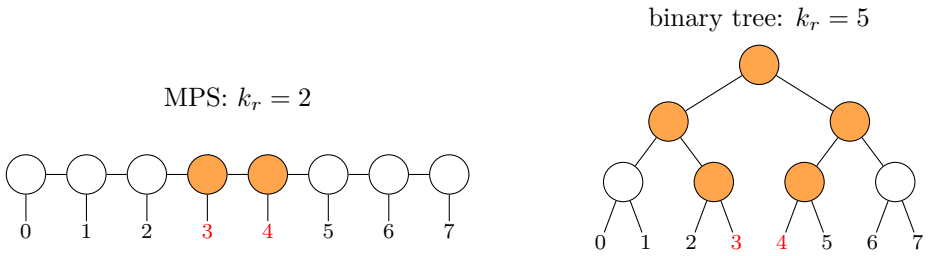



Figure 1: Learned component $\pi_j = 00000000$, unlearned $\pi_r = 00011000$. They differ at sites 3 and 4 (red). Orange: the Steiner subtree $\mathrm{St}(V_r)$. On the MPS the two sites are on adjacent cores ($k_r = 2$); on the binary tree the subtree passes through the root ($k_r = 5$); on the HT tree $k_r = 7$.

## 3 Two ways to decrease the loss

Before stating the theorem we describe the two kinds of perturbation that lower the loss at $\theta_S$. The theorem says that nothing else does better.

**New channel along a subtree.** An entry on an unused channel of edge $e$ is multiplied, in the contraction, by the neighbouring core's entries on the same channel, which are zero at $\theta_S$. So it changes $W$ only if the neighbour also uses that channel. A set of cores therefore changes $W$ through unused channels only if it is connected and all its cores use them together; the change is then a product of one small entry per core. To produce component $r$ in this way, the set must contain every core on which $\pi_r$ differs from $\pi_j$, hence all of $\mathrm{St}(V_r)$. With entries of size $\tau$ on the $k_r$ cores of $\mathrm{St}(V_r)$,

$$L - L_S = -c_r\,\tau^{k_r} + O(\tau^{2k_r}), \qquad c_r = s_r\,s_j^{(K-k_r)/K} \tag{5}$$

(Proposition 1, at the balanced point). So the order is at most $k^*$.

**Combining two cheaper perturbations.** A perturbation can also change the indices of $W$ on a set $B$ of cores at lower cost, in two ways:

- if $B = \{v\}$ is a single core, by rotating its vector $q_v$ toward another basis vector, keeping the used channel; the change in $W$ is linear in the rotation angle;
- if $|B| \geq 2$, by a new channel along $\mathrm{St}(B)$ as above; the change in $W$ has size $\eta$ when each of the $|\mathrm{St}(B)|$ entries has size $\eta^{1/|\mathrm{St}(B)|}$ (in units of $a$).

Each such perturbation moves $W$ away from $s_j U_j$, which costs $\frac{1}{2}s_j^2\eta^2$ at second order. Two of them on disjoint core sets $B_1, B_2$ also produce, at second order, the product tensor with modified indices on $B_1 \cup B_2$. If that tensor is an unlearned component $r$, this gains $s_r s_j \eta_1 \eta_2$. The net change is a quadratic form in the amplitudes, which is negative in some direction iff the gains beat the costs. The matrix below records the gains.

**Definition 1** (Coupling matrix)**.** For $\ell \geq 1$, let $\mathcal{N}_\ell$ be the set of pairs $(B, \beta)$, where $B$ is a nonempty set of cores with $|\mathrm{St}(B)| \leq \ell$ and $\beta$ assigns a basis index to each site of $B$ such that, on every core of $B$, the indices differ from those of $\pi_j$. The coupling matrix $M_\ell$ is the symmetric matrix indexed

by $\mathcal{N}_\ell$ with entries

$$M_\ell[(B_1,\beta_1),(B_2,\beta_2)] = \begin{cases} s_r & \text{if } B_1 \cap B_2 = \emptyset \text{ and the string equal to } \beta_1 \text{ on } B_1,\ \beta_2 \text{ on } B_2, \\ & \text{and } \pi_j \text{ elsewhere is } \pi_r \text{ for some component } r, \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

Let $\mu_\ell$ be its largest eigenvalue, and

$$\ell^* = \min\{\ell \geq 1 :\ \mu_\ell > s_j\} \qquad (\ell^* = \infty \text{ if there is no such } \ell). \tag{7}$$

An unordered pair of elements determines one string, hence at most one component, so each unordered pair contributes one weight $s_r$ and $\frac{1}{2}x^\top M_\ell x = \sum_{\{n_1,n_2\}} M_\ell[n_1,n_2]x_{n_1}x_{n_2}$, summed over unordered pairs. $M_\ell$ is symmetric with nonnegative entries, so $\mu_\ell$ is also its spectral radius (Perron–Frobenius), and $\mu_\ell$ is nondecreasing in $\ell$ because $M_\ell$ is a principal submatrix of $M_{\ell+1}$. A nonzero entry requires a component $r$ with $V_r = B_1 \sqcup B_2$, so if no $V_r$ splits into two sets whose Steiner subtrees have at most $\ell$ cores, then $M_\ell = 0$.

At level $\ell = 1$ only single-core rotations occur. Let $z_u$ be the entry of $G_u$ with a physical index other than that of $q_u$ and all edge indices 0. The block of the Hessian at $\theta_S$ in these coordinates is

$$H_{\text{rot}} = s_j\, D(s_j I - M_1)D, \qquad D = \operatorname{diag}(1/a_u), \tag{8}$$

which is $s_j^{(K-2)/K}(s_j I - M_1)$ at the balanced point. Since $D$ is invertible, $H_{\text{rot}}$ has a negative eigenvalue iff $\mu_1 > s_j$. For $\ell \geq 2$ the combined perturbation has $\|\theta - \theta_S\| \sim \eta^{1/\ell}$ and lowers the loss by $\sim \eta^2$, which is order $2\ell$ in the distance.

## 4 Main result

**Theorem 1** (Order of the saddle after one learned component)**.** *Let $W^*$ be as in (2) and $\theta_S$ as in (3), with $\mathcal{R} \neq 0$ and $k^* \geq 2$. Assume*

*(a) $\mu_\ell \neq s_j$ for every $\ell < k^*/2$;*

*(b) if $2\ell^* \geq k^*$: $\chi_e \geq 2$ on the edges of $\mathrm{St}(V_r)$ for some $r$ with $k_r = k^*$;*
*if $2\ell^* < k^*$: there is a set $X \subseteq \mathcal{N}_{\ell^*}$ such that the submatrix of $M_{\ell^*}$ on $X$ has largest eigenvalue above $s_j$ and every edge $e$ has $\chi_e - 1 \geq \#\{(B,\beta) \in X : |B| \geq 2,\ e \text{ is an edge of } \mathrm{St}(B)\}$.*

*Then*

$$\operatorname{ord}(\theta_S) = \min(k^*,\, 2\ell^*).$$

Assumption (a) excludes boundary cases. Assumption (b) provides the unused channels needed by the descent of order $k^*$ (first case) or $2\ell^*$ (second case). The theorem follows from three bounds, proved in Section 7: ord $\leq k^*$ (Proposition 1); ord $\geq \min(k^*, 2\ell)$ whenever $\mu_{\ell-1} < s_j$ (Proposition 2); and ord $\leq 2\ell$ whenever $\mu_\ell > s_j$ and $\ell < k^*/2$ (Proposition 3).

When $2\ell^* < k^*$, the descent of order $2\ell^*$ moves toward index strings that are not components of the target: each element $(B,\beta)$ it uses with $|B| \geq 2$ has $|\mathrm{St}(B)| \leq \ell^* < k^*$, while producing a component alone would require $B = V_r$ and hence $|\mathrm{St}(B)| \geq k^*$. Components appear only through pairs of elements.

**Remark 1** (General targets)**.** The lower bound does not need (2). For any $W^*$ and unit vectors $q_v$, let $U_j = \bigotimes_v q_v$, $s_j = \langle W^*, U_j\rangle > 0$, $\mathcal{R} = W^* - s_j U_j$, and $\theta_S$ as in (3). Then $\theta_S$ is critical iff $P_{\{v\}}\mathcal{R} = 0$ for every core $v$ (notation of Section 7), and Proposition 2 holds with the condition $\lambda_{\max}(S_{\ell-1}) < s_j$ on the operator defined there, which reduces to $\mu_{\ell-1} < s_j$ for targets (2).

**Corollary 1.** *Under the assumptions of Theorem 1:*

*(i) If $k^* = 2$, the saddle is strict.*

*(ii) If $k^* \in \{3, 4\}$, the order is $k^*$ if $\mu_1 < s_j$ and 2 if $\mu_1 > s_j$. So the order is $k^*$ unless the Hessian has a negative eigenvalue.*

*(iii) If $W^* = s_j U_j + s_r U_r$ with $s_r < s_j$ (the larger component was learned), the order is $k_r$.*

*(iv) If $W^* = s_j U_j + s_r U_r$ with $s_r > s_j$, let $\ell_0 = \min \max(|\mathrm{St}(B_1)|, |\mathrm{St}(B_2)|)$ over partitions of $V_r$ into two nonempty sets $B_1, B_2$. The order is $\min(k_r, 2\ell_0)$.*

*(v) The order is below $k^*$ while the Hessian is positive semidefinite only if $k^* \geq 5$.*

Part (iii) covers the typical situation in training, where components are learned in decreasing order of weight: then the order is $k_r$, the size of a Steiner subtree, which depends on the tree.

**Example 1** (Order below $k^*$ with a positive semidefinite Hessian)**.** MPS with six cores, $d = \chi = 2$, learned $\pi_j = 000000$ with $s_j = 1$, and two unlearned components $\pi_1 = 110001$ and $\pi_2 = 110010$ with weight 0.8 each. Here $V_1 = \{0, 1, 5\}$, so $k_1 = 6$, and $V_2 = \{0, 1, 4\}$, so $k_2 = 5$; hence $k^* = 5$. No component differs from $\pi_j$ on exactly two cores, so $M_1 = 0$. In $M_2$, the element $(\{0, 1\}, 11)$ is coupled to $(\{5\}, 1)$ through component 1 and to $(\{4\}, 1)$ through component 2, so $\mu_2 = 0.8\sqrt{2} \approx 1.13 > s_j$ and $\ell^* = 2$. Theorem 1 gives order $\min(5, 4) = 4$. The Hessian is positive semidefinite: $\mu_1 = 0 < s_j$, so Proposition 2 with $\ell = 2$ gives order $\geq 4 > 2$.

The descent uses a new channel on cores 0 and 1 with entries $u, w$ that produce the indices 11 there (the string 110000, not a component), together with rotations $\alpha, \beta$ of cores 4 and 5. With $y = uw$ and in units $s_j = 1$, $L - L_S = \frac{1}{2}(y^2 + \alpha^2 + \beta^2) - 0.8\, y(\alpha + \beta)$ plus higher-order terms. Minimizing over $\alpha, \beta$ gives $\alpha = \beta = 0.8y$ and $L - L_S = -0.14\, y^2$. On the sphere $u^2 + w^2 = \tau^2$, $y \leq \tau^2/2$, so this family reaches $-0.035\,\tau^4 = -(2 \cdot 0.8^2 - 1)\tau^4/8$. Numerically, the minimum of $L - L_S$ over all of the sphere $\|\theta - \theta_S\| = \tau$ has slope 4.000 in $\tau$ and coefficient $-0.0350$, so nothing else contributes at order $\tau^4$. With weights 0.6 instead, $\mu_2 = 0.6\sqrt{2} \approx 0.85 < 1$ and the order is 5.

## 5 Plateau lengths

The following is a leading-order derivation, not a theorem.

**Escape time.** Start gradient flow at the balanced $\theta_S$ plus a perturbation of size $\delta$ along an order-$p$ descent of Section 3. At leading order in $\delta$, with the other variables relaxed to their instantaneous minimum, the amplitude $b$ of the descent obeys $\dot{b} = c\, b^{p-1}$ with $c > 0$, so the time to leave the saddle is

$$t_{\mathrm{esc}} \simeq \begin{cases} \ln(1/\delta)/c, & p = 2, \\ \delta^{-(p-2)}/((p-2)\, c), & p \geq 3. \end{cases} \tag{9}$$

This is derived (Section 7.6) in three cases: a new channel along $\mathrm{St}(V_r)$ with equal entries ($p = k_r$, $c = c_r = s_r s_j^{(K-k_r)/K}$); pure rotations ($\ell^* = 1$, $p = 2$), where the escape is a linear instability with

rate $c = -\lambda_{\min}(H_{\rm rot})$; and the descent of order $2\ell^*$ when $X$ contains a single element with $|B| \geq 2$. When $2\ell^* < k^*$ both descents are present, and the lower order governs the escape only for $\delta$ below a crossover scale set by their coefficients.

**From small initialization.** In a run from initialization scale $\sigma$, the entries that will grow are of order $\sigma$ when the flow reaches $\theta_S$, so the plateau there lasts $\propto \ln(1/\sigma)$ or $\propto \sigma^{-(p-2)}$. These entries already grow while component $j$ is being learned. For $W^* = s_j U_j + s_r U_r$ with $s_r < s_j$, accounting for this growth gives the time $\Delta t$ between learning $j$ and learning $r$:

$$\Delta t \simeq \frac{1 - s_r/s_j}{s_r s_j^{(K-2)/K}} \ln \frac{1}{\sigma} \quad (k_r = 2), \qquad \Delta t \simeq \frac{1 - s_r/s_j}{(k_r - 2)\, s_r s_j^{(K-k_r)/K}} \, \sigma^{-(k_r-2)} \quad (k_r \geq 3), \tag{10}$$

where the second formula assumes an initialization aligned with the target's basis vectors.

**Evidence.**

- *Order.* Minimizing $L - L_S$ numerically over spheres of radius $\tau$ in 32 cases (MPS, binary and HT trees, $k^*$ from 2 to 7, including targets not of the form (2)) gives slopes within 0.003 of the order predicted by Theorem 1 in all 30 cases it covers; 12 further cases computed with independent code agree to $5 \cdot 10^{-4}$.
- *Two components, $s_r < s_j$.* In about 11,600 gradient-flow runs on nine trees ($K$ from 3 to 15), the fitted exponent of $\Delta t$ in $1/\sigma$ is $k_r - 2$ for $k_r = 3, 4, 5$, and for $k_r = 6, 7$ on two trees with balanced initialization. For $k_r = 2$ a logarithmic law fits better than a power law in all 14 settings, with slope 0.98–1.01 against the value 1 predicted by (10). On every edge outside $\mathrm{St}(V_r)$ the second singular value of the corresponding matricization of $W$ stays below $4.3\,\sigma^2 s_r$.
- *Order $2\ell^* < k^*$.* A pre-registered test (about 2,800 runs) started gradient flow at $\theta_S$ plus a perturbation of size $\delta$ and fitted $t_{\rm esc} \propto \delta^{-\gamma}$. For $\delta$ below the predicted crossover it found $\gamma = 2.001 \pm 0.001$ for $p = 4$ ($k^* = 5$ and 6) and $\gamma = 3.996$–$3.998$ for $p = 6$ ($k^* = 7$); the exponent $k^* - 2$ fits $10^3$–$10^5$ times worse. Above the crossover the results were mixed. This test has not been independently re-checked.

## 6 Relation to prior work

**Deep linear networks.** Saddle-to-saddle learning, the exponent $K - 2$ and learning in order of decreasing singular value are established for DLNs (Saxe et al., 2014; Jacot et al., 2022). Achour et al. (2024, Thm 7) show which DLN saddles are strict, a lower bound of the form "order $\geq 3$". A binary tree with four sites is exactly a depth-3 DLN; there $k^* = 3$ and Theorem 1 reduces to that classification.

**Homogeneous models.** Near the origin the weights of a homogeneous model align with a critical direction of the correlation with the target and escape after a time $\propto \sigma^{-(K-2)}$ (Kumar and Haupt, 2025). If 1, 2 or $k \geq 3$ layers start small while the others are large, the escape takes time $O(1)$, $\propto \ln(1/\sigma)$ or $\propto \sigma^{-(k-2)}$ (Rawal and DeWeese, 2026, Thm 4.2). Small weights keep the growth they acquired during earlier plateaus (Kunin et al., 2025; Zhang et al., 2026), which is the factor $1 - s_r/s_j$ in (10). These results take the number of small factors as given; Theorem 1 determines it for TTNs.

**The landscape of TTNs.** Furman et al. (2026) analyse the landscape of the same model and loss. They prove that local minima of minimum norm are global. At a minimum-norm critical point they consider the edges on which the rank of the model is below that of the target. Each connected set of such edges, touching $n$ cores, gives a straight-line perturbation along which the loss decreases at order $n$ (their Lemmas D.8–D.9), an upper bound on the order. They remark that with several separate such sets the smallest gives the fastest escape. They prove a lower bound for one saddle (a parity target, their Prop. 5.3) and do not study the dynamics.

**What is new.**

(1) A lower bound on the order at the saddles (3), matching the upper bound, so the order is determined exactly. Previously lower bounds were known only for DLNs and one parity saddle.

(2) The order is set by the smallest single Steiner subtree $\mathrm{St}(V_r)$. At $\theta_S$ an edge has model rank below target rank exactly when cores of $\bigcup_r V_r$ lie on both sides of it, so the set of such edges is the edge set of $\mathrm{St}(\bigcup_r V_r)$. For two components this is $\mathrm{St}(V_r)$, the upper bound of Furman et al. (2026) already equals $k_r$, and the new part is the lower bound. With more components, $\mathrm{St}(\bigcup_r V_r)$ can be much larger than the smallest $\mathrm{St}(V_r)$, which sets the order.

(3) The case $\mu_\ell > s_j$: the order $2\ell^*$, and descent through index strings that are not components of the target.

(4) The escape laws (9)–(10) for TTNs and their numerical tests.

# 7 Proofs

Throughout, $\varepsilon = \|\theta - \theta_S\|$, $\nu_v$ is the vector of entries of core $v$ on unused channels (entries with at least one nonzero edge index), and $C_0, C_1, C_2, C_\gamma, M$ denote constants that do not depend on $\varepsilon$. Sets of cores in this section are denoted $A, A_1, A_2, B$.

## 7.1 Decomposition by the cores on which a tensor differs from $U_j$

Let $p_v = q_v q_v^\top$ and $p_v^\perp = I - p_v$ on $\mathbb{R}^{D_v}$, and for a set $B$ of cores

$$P_B = \bigotimes_{v \in B} p_v^\perp \otimes \bigotimes_{v \notin B} p_v,$$

with tensor factors labelled by cores. The $2^K$ projections $P_B$ are orthogonal and sum to the identity. Let $H_B = \bigotimes_{v \in B} q_v^\perp$ (factors labelled by the cores of $B$); it embeds in $\bigotimes_v \mathbb{R}^{D_v}$ by tensoring with $q_v$ for $v \notin B$, and its image is the range of $P_B$, the tensors that differ from $U_j$ exactly on the cores of $B$. A basis of $H_B$ is given by the products of basis vectors whose indices differ from those of $\pi_j$ on every core of $B$. For disjoint $B_1, B_2$, $H_{B_1} \otimes H_{B_2} = H_{B_1 \sqcup B_2}$, with factors matched by core label. Let $\mathcal{R}_B = P_B \mathcal{R}$, viewed in $H_B$. By (2),

$$\mathcal{R}_B = \sum_{r:\, V_r = B} s_r U_r, \qquad \text{so} \qquad k^* = \min\{|\mathrm{St}(B)| : \mathcal{R}_B \neq 0\}. \tag{11}$$

**Lemma 1.** *(i) $\theta_S$ is critical iff $\mathcal{R}_{\{v\}} = 0$ for every core $v$, i.e. iff $k^* \geq 2$. (ii) Let $\omega(\theta) = \langle U_j, W(\theta)\rangle$ and, for $B \neq \emptyset$, $m_B = P_B W/\omega \in H_B$. Wherever $\omega \neq 0$,*

$$L(\theta) - L_S = \tfrac{1}{2}(\omega - s_j)^2 + \sum_{B\neq\emptyset} \Big[\tfrac{\omega^2}{2}\|m_B\|^2 - \omega\langle \mathcal{R}_B, m_B\rangle\Big]. \tag{12}$$

*Proof.* (i) Entries on unused channels do not change $W$ at first order (Section 3). A change of the entries of core $v$ on the used channel changes $W$ within $H_\emptyset \oplus H_{\{v\}}$. So the gradient vanishes iff $\mathcal{R}$ has no component in any $H_{\{v\}}$; its component in $H_\emptyset$ is zero because $\langle U_j, \mathcal{R}\rangle = 0$. (ii) By orthogonality, $\|W - W^*\|^2 = \sum_B \|P_B W - P_B W^*\|^2$, with $P_\emptyset(W - W^*) = (\omega - s_j)U_j$, and for $B \neq \emptyset$, $P_B W^* = \mathcal{R}_B$ and $P_B W = \omega m_B$. Also $L_S = \frac{1}{2}\sum_B \|\mathcal{R}_B\|^2$. □

### 7.2 Upper bound $k^*$

**Proposition 1.** *Let $k_r = k^*$ and $\chi_e \geq 2$ on the edges of $\mathrm{St}(V_r)$. At the balanced point, set the entry of each core $v \in \mathrm{St}(V_r)$ with physical index given by $\pi_r$ on the sites of $v$, index 1 on the edges of $\mathrm{St}(V_r)$ and index 0 on its other edges, to $\tau$. Then $\|\theta(\tau) - \theta_S\| = \sqrt{k^*}\,\tau$ and*

$$L(\theta(\tau)) - L_S = -s_r\, s_j^{(K-k^*)/K}\,\tau^{k^*} + \tfrac{1}{2} s_j^{2(K-k^*)/K}\,\tau^{2k^*}.$$

*Proof.* Expand the contraction by choosing, for each core of $\mathrm{St}(V_r)$, either its original entry or the new one. If a term uses the new entry on one core of $\mathrm{St}(V_r)$ and the original entry on a neighbouring core of $\mathrm{St}(V_r)$, the shared edge contracts index 1 with index 0 and the term vanishes. Only two terms remain: all original entries, giving $s_j U_j$, and all new entries on $\mathrm{St}(V_r)$, giving $\tau^{k^*} a^{K-k^*} U$, where $U$ has the indices of $\pi_r$ on the sites of $\mathrm{St}(V_r)$ and those of $\pi_j$ elsewhere. Since $\pi_r = \pi_j$ outside $V_r$, $U = U_r$. Hence $W - W^* = -\mathcal{R} + \tau^{k^*} a^{K-k^*} U_r$, and $\langle \mathcal{R}, U_r\rangle = s_r$. □

### 7.3 Lower bound

**Cumulants.** Consider families $f = (f_B)_{B\subseteq V}$ with $f_B \in H_B$, with the product $(f * g)_B = \sum_{B_1 \sqcup B_2 = B} f_{B_1} \otimes g_{B_2}$. Because tensor factors are labelled by core, this product is commutative and associative, with unit $(1, 0, 0, \dots)$, and every family with $f_\emptyset = 0$ is nilpotent; so exp and log are finite sums and $\log(f * g) = \log f + \log g$. With $m_\emptyset = 1$, define $\kappa = \log m$. Explicitly,

$$m_B = \sum_{\mathcal{P}\in \mathrm{Part}(B)} \bigotimes_{A\in\mathcal{P}} \kappa_A, \qquad \kappa_B = \sum_{\mathcal{P}\in\mathrm{Part}(B)} (-1)^{|\mathcal{P}|-1}(|\mathcal{P}|-1)! \bigotimes_{A\in\mathcal{P}} m_A,$$

sums over set partitions $\mathcal{P}$ of $B$. Near $\theta_S$, $\omega$ is close to $s_j \neq 0$, so the $m_B$ and $\kappa_B$ are analytic there; they vanish at $\theta_S$ for $B \neq \emptyset$, so $\|\kappa_B\| \leq C_0\varepsilon$.

**Lemma 2.** *For $|B| \geq 2$, near $\theta_S$, $\|\kappa_B\| \leq C_0 \prod_{x\in\mathrm{St}(B)} \|\nu_x\|$. In particular $\|\kappa_B\| \leq C_0 \varepsilon^{|\mathrm{St}(B)|}$.*

*Proof. Vanishing.* Fix $x \in \mathrm{St}(B)$ and suppose $\nu_x = 0$. Then all nonzero entries of $G_x$ have edge indices 0, so $W$ factorizes at $x$: $W = g_x \otimes \bigotimes_b W_b$, where $g_x \in \mathbb{R}^{D_x}$ is the physical vector of $G_x$, $b$ runs over the components of $T$ with $x$ removed (node sets $V_b$), and $W_b$ is the contraction of component $b$ with its edge to $x$ set to index 0. Correspondingly $\omega = \omega_x \prod_b \omega_b$ with $\omega_x = \langle q_x, g_x\rangle$ and $\omega_b = \langle \bigotimes_{v\in V_b} q_v, W_b\rangle$, all close to their nonzero values at $\theta_S$. The projections $P_B$ factorize core by core, so $m = m^{(x)} * \prod_b m^{(b)}$, where $m^{(x)}_{\{x\}} = p_x^\perp g_x/\omega_x$, $m^{(b)}_A = P^{(b)}_A W_b/\omega_b$ for $A \subseteq V_b$ (with $P^{(b)}_A$ the analogous projection on the sites of $V_b$), and each has unit component on $\emptyset$. Hence

$\kappa = \log m^{(x)} + \sum_b \log m^{(b)}$ is supported on subsets of $\{x\}$ or of a single $V_b$. $B$ is neither: if $x \in B$, then $B \not\subseteq \{x\}$ and $B \not\subseteq V_b$; if $x \notin B$ and $B \subseteq V_b$, then the connected set $V_b$ contains $\mathrm{St}(B)$, which contains $x$. So $\kappa_B = 0$ whenever $\nu_x = 0$ for some $x \in \mathrm{St}(B)$.

*Bound.* Expand $\kappa_B$ in a power series in $\theta - \theta_S$, absolutely convergent on a polydisc of radius $2r_0$. Setting $\nu_x = 0$ removes exactly the monomials containing a variable of $\nu_x$, and the result is 0; so every monomial with nonzero coefficient contains a variable of $\nu_x$, for each $x \in \mathrm{St}(B)$. Fix an ordering of the variables, and from each monomial factor out, for each $x \in \mathrm{St}(B)$, the first variable of $\nu_x$ that it contains. The remaining series converges on the polydisc of radius $r_0$ and is bounded there, which gives the claim. □

**Proposition 2.** *Let $\theta_S$ be critical and $2 \le \ell \le \lceil k^*/2 \rceil$. On $\mathcal{H}_{\ell-1} = \bigoplus H_A$, summed over nonempty $A$ with $|\mathrm{St}(A)| \le \ell - 1$, define the symmetric bilinear form*

$$Q(x, y) = \sum_{A_1 \cap A_2 = \emptyset} \langle \mathcal{R}_{A_1 \sqcup A_2}, x_{A_1} \otimes y_{A_2} \rangle,$$

*and let $S_{\ell-1}$ be its operator. If $\lambda_{\max}(S_{\ell-1}) < s_j$, then $L(\theta) - L_S \ge -C_1 \varepsilon^{\min(k^*, 2\ell)}$ near $\theta_S$. For targets (2), $S_{\ell-1} = M_{\ell-1}$ in the basis of products of basis vectors.*

*Proof.* Let $p = \min(k^*, 2\ell)$ and $y_A = \|\kappa_A\|$. Call $A$ *small* if $|\mathrm{St}(A)| \le \ell - 1$ and *large* otherwise. Then: (F1) a large set has $|\mathrm{St}(A)| \ge \ell \ge 2$, so $y_A \le C_0 \varepsilon^\ell$ by Lemma 2; (F2) subsets of small sets are small; (F3) $\mathcal{R}_A = 0$ for small $A$, since $|\mathrm{St}(A)| < k^*$; (F4) summing $\langle \mathcal{R}_B, \kappa_{A_1} \otimes \kappa_{A_2} \rangle$ over all $B$ and all unordered partitions $B = A_1 \sqcup A_2$ into small sets gives $\frac{1}{2} Q(\kappa_{\mathrm{s}}, \kappa_{\mathrm{s}}) \le \frac{1}{2} \lambda_{\max}(S_{\ell-1}) \|\kappa_{\mathrm{s}}\|^2$, where $\kappa_{\mathrm{s}}$ collects the cumulants of small sets.

*Quadratic term of (12).* For small $B$, $m_B - \kappa_B$ is a sum of products of at least two cumulants of small sets (F2), each at most $C_0 \varepsilon\, y_A$ for one factor $A$. So $\sum_{B \text{ small}} \|m_B\|^2 \ge (1 - C_2 \varepsilon) \|\kappa_{\mathrm{s}}\|^2$; the terms with large $B$ are nonnegative and dropped.

*Linear term.* Only $B$ with $\mathcal{R}_B \neq 0$ contribute, and they have $|\mathrm{St}(B)| \ge k^*$. Expanding $m_B$ over set partitions: the single cumulant gives $O(\varepsilon^{k^*})$ (Lemma 2); two small blocks give (F4); one small and one large block give at most $\frac{\gamma}{2} y_{A_1}^2 + C_\gamma \varepsilon^{2\ell}$ (Young's inequality and F1); two large blocks give $O(\varepsilon^{2\ell})$; three or more blocks give $O(\varepsilon)(y_{A_1}^2 + y_{A_2}^2)$ if two blocks are small and $O(\varepsilon^{2\ell})$ otherwise. With $M$ the number of terms in which a given $y_A^2$ appears, $\sum_B \langle \mathcal{R}_B, m_B \rangle \le \frac{1}{2}(\lambda_{\max}(S_{\ell-1}) + M\gamma + C_0' \varepsilon) \|\kappa_{\mathrm{s}}\|^2 + C_\gamma \varepsilon^p$.

*Conclusion.* By (12), dropping $\frac{1}{2}(\omega - s_j)^2$, $L - L_S \ge \frac{\omega}{2}[\omega(1 - C_2\varepsilon) - \lambda_{\max}(S_{\ell-1}) - M\gamma - C_0' \varepsilon] \|\kappa_{\mathrm{s}}\|^2 - \omega C_\gamma \varepsilon^p$. With $g = s_j - \lambda_{\max}(S_{\ell-1}) > 0$, $\gamma = g/(4M)$ and $\omega \ge s_j - C_0'' \varepsilon$, the bracket is positive for small $\varepsilon$, so $L - L_S \ge -C_1 \varepsilon^p$. □

### 7.4 Upper bound $2\ell$

**Proposition 3.** *Let $\ell < k^*/2$ and let $X \subseteq \mathcal{N}_\ell$ be such that $M_\ell$ restricted to $X$ has largest eigenvalue $\rho > s_j$ and $X$ satisfies the edge condition of Theorem 1(b). Then some curve has $\|\theta(\tau) - \theta_S\| \le \tau$ and $L(\theta(\tau)) - L_S \le -c\tau^{2\ell}$ for some $c > 0$.*

*Proof.* Work at the balanced point. Let $x \ge 0$ be a unit eigenvector of $M_\ell|_X$ for $\rho$; it exists by Perron–Frobenius since the entries are nonnegative, and $x \ge 0$ makes the roots below real. For $0 < \eta \le 1$:

- for $(\{v\}, \beta) \in X$, set the entry of $G_v$ with physical index $\beta$ and all edge indices 0 to $a\eta x_{(\{v\},\beta)}$ (a rotation of $q_v$);

- for $n = (B, \beta) \in X$ with $|B| \geq 2$, let $h = |\mathrm{St}(B)| \leq \ell$. On every edge of $\mathrm{St}(B)$ choose an unused index chosen by no other such element of $X$ (possible by the edge condition). Give each core $u \in \mathrm{St}(B)$ the entry $a(\eta x_n)^{1/h}$ with physical index $\beta$ (if $u \in B$) or that of $\pi_j$ (if $u \notin B$), the chosen indices on the edges of $\mathrm{St}(B)$, and 0 on its other edges.

Then $\|\theta - \theta_S\| \leq C_0 \eta^{1/\ell}$.

*Structure of $W$.* Consider a nonzero term of the contraction and an edge $e$ carrying a nonzero index. That index was chosen by exactly one element $n$, so both cores at $e$ use entries of $n$. The entries of $n$ carry $n$'s indices on every edge of $\mathrm{St}(B)$ and 0 on all other edges, so the term uses $n$'s entries on all of $\mathrm{St}(B)$ and on no core outside it. Hence each connected set of edges with nonzero indices is the edge set of $\mathrm{St}(B)$ for a single element, used completely. $W$ is therefore a sum over collections of elements of $X$ on pairwise disjoint cores, in which each element contributes the factor $\eta x_n$ (the $h$ roots multiply to $\eta x_n$) and changes the indices on its set $B$, and every other core contributes $a$ times its original vector. In particular $W$ is a polynomial in $\eta$ with integer powers, $\omega = s_j$ exactly, and $m_B = \eta\, m_B^{(1)} + \eta^2 m_B^{(2)} + O(\eta^3)$, where $m^{(1)}$ collects single elements and $m^{(2)}$ pairs of elements.

*Order $\eta$.* $\sum_B \|m_B^{(1)}\|^2 = \|x\|^2 = 1$. No single element has $B = V_r$, since that would need $|\mathrm{St}(B)| \geq k^* > \ell$; so $\sum_B \langle \mathcal{R}_B, m_B^{(1)} \rangle = 0$ by (11).

*Order $\eta^2$.* If $M_\ell[n_1, n_2] = s_r > 0$, then $\mathrm{St}(B_1)$ and $\mathrm{St}(B_2)$ are disjoint: otherwise their union would be a connected set containing $V_r$ with at most $2\ell - 1 < k^*$ cores. So every nonzero entry of $M_\ell|_X$ corresponds to a pair of elements on disjoint cores, and $\sum_B \langle \mathcal{R}_B, m_B^{(2)} \rangle = \frac{1}{2} x^\top M_\ell|_X x = \frac{\rho}{2}$.

By (12) with $\omega = s_j$: $L - L_S = \frac{s_j^2}{2}\eta^2 - \frac{s_j \rho}{2}\eta^2 + O(\eta^3) = -\frac{s_j}{2}(\rho - s_j)\eta^2 + O(\eta^3)$. Setting $\eta = (\tau / C_0)^\ell$ gives the claim. □

### 7.5 Proof of Theorem 1 and Corollary 1

If $k^* = 2$, the curve of Proposition 1 has order 2, every critical point has order at least 2, and $\min(2, 2\ell^*) = 2$. Let $k^* \geq 3$.

*Case $2\ell^* \geq k^*$* (including $\ell^* = \infty$). Every $\ell < k^*/2$ has $\ell < \ell^*$, so $\mu_\ell \leq s_j$, and $\mu_\ell < s_j$ by (a). Proposition 2 with $\ell = \lceil k^*/2 \rceil$ gives ord $\geq k^*$, and Proposition 1 gives ord $\leq k^*$. So ord $= k^* = \min(k^*, 2\ell^*)$.

*Case $2\ell^* < k^*$.* If $\ell^* = 1$, Proposition 3 gives ord $\leq 2$, so ord $= 2$. If $\ell^* \geq 2$, then $\mu_{\ell^*-1} \leq s_j$ by minimality, and $< s_j$ by (a); Proposition 2 at $\ell = \ell^* \leq \lceil k^*/2 \rceil$ gives ord $\geq \min(k^*, 2\ell^*) = 2\ell^*$, and Proposition 3 gives ord $\leq 2\ell^*$. □

*Corollary.* (i) Shown above. (ii) An order below $k^* \leq 4$ requires $2\ell^* < 4$, i.e. $\ell^* = 1$, i.e. $\mu_1 > s_j$. For (8): changing $z_u$ changes $W$ by $z_u\, s_j / a_u$ times an orthonormal tensor, and two such coordinates on cores $u, v$ couple through any component with $V_r = \{u, v\}$, with coefficient $-s_r\, s_j / (a_u a_v)$. So the block is $s_j D(s_j I - M_1) D$. (iii) With one unlearned component, each row of $M_\ell$ has at most one nonzero entry, $s_r$, so $\mu_\ell \leq s_r < s_j$ for all $\ell$ and $\ell^* = \infty$. (iv) Each row again has at most one nonzero entry, and some entry is nonzero iff $\ell \geq \ell_0$; so $\mu_\ell = s_r > s_j$ for $\ell \geq \ell_0$, $\ell^* = \ell_0$, and the order is $\min(k_r, 2\ell_0)$. (v) For $k^* \leq 4$, by (ii) an order below $k^*$ means order 2, which is equivalent to a negative Hessian eigenvalue (Section 2). □

### 7.6 Derivation of the escape times

*New channel along* $\mathrm{St}(V_r)$. With the other entries fixed at the balanced $\theta_S$ and amplitudes $b_v$ on the cores of $\mathrm{St}(V_r)$, the computation of Proposition 1 gives $\dot{b}_v = a^{K-k_r} \prod_{u \neq v} b_u \, (s_r - a^{K-k_r} \prod_u b_u)$. For equal small amplitudes, $\dot{b} = c_r b^{k_r - 1}$, and $\frac{d}{dt} b^{-(p-2)} = -(p-2)c$ integrates to (9).

*Rotations ($\ell^* = 1$).* The flow is linear at leading order, $\dot{z} = -H_{\mathrm{rot}} z$, and grows along the eigenvector of the most negative eigenvalue of $H_{\mathrm{rot}}$ at rate $c = -\lambda_{\min}(H_{\mathrm{rot}})$.

*Order $2\ell^*$ with one multi-core element.* Let $X$ contain one element $(B, \beta)$ with $|B| \geq 2$, and $h = |\mathrm{St}(B)|$. Then $h = \ell^*$: if $h < \ell^*$, all elements of $X$ would lie in $\mathcal{N}_{\ell^*-1}$, so $\mu_{\ell^*-1} \geq \rho > s_j$, contradicting the minimality of $\ell^*$. The rotations have curvature of order one and follow the amplitude $y = \prod_{v \in \mathrm{St}(B)} b_v$; minimizing over them leaves $L - L_S \approx -\frac{c}{2} y^2$ with $c > 0$ (in Example 1, $c = 2(0.14) = 0.28$). Then $\dot{b}_v = c \, b_v \prod_{u \neq v} b_u^2$, and for equal amplitudes $\dot{b} = c \, b^{2h-1} = c \, b^{2\ell^*-1}$.

*Growth before the plateau.* While $j$ is learned from aligned initialization, its cores grow as $\dot{a} \simeq s_j a^{K-1}$ from $a = \sigma$, and the entries on $\mathrm{St}(V_r)$ already grow at rate $s_r a^{K-k_r} b^{k_r-1}$. For $k_r \geq 3$, $b^{-(k_r-2)}$ decreases by $(k_r - 2) s_r \int a^{K-k_r} dt = (k_r - 2) \frac{s_r}{s_j} \int_\sigma a^{1-k_r} da = \frac{s_r}{s_j} \sigma^{-(k_r-2)} + O(1)$, so at the plateau $b^{-(k_r-2)} = (1 - s_r/s_j) \sigma^{-(k_r-2)} + O(1)$. For $k_r = 2$, $\ln b$ grows by $\frac{s_r}{s_j} \ln \frac{1}{\sigma} + O(1)$. Inserting into (9) gives (10).

## 8 Scope

The theorem covers the saddles (3) with one learned component, targets (2) for the exact value (the lower bound holds for any target, Remark 1), whitened inputs and the population loss. Saddles with several learned components and the boundary cases $\mu_\ell = s_j$ are open. Section 5 is a leading-order derivation, not a theorem, and covers the three cases listed there.

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